Part A
Logical Interpretation

1. Title of the Research Project

Hybrid KD-Guided, Adapter-Based Diffusion-Transformer for Low-Light and Uncertainty-Aware Diabetic Retinopathy Detection on Edge IoT Devices

2. Statement of the Problem

Diabetic Retinopathy (DR) is one of the leading causes of vision loss worldwide, particularly among diabetic patients. Early detection of DR is essential to prevent severe complications. In recent years, deep learning-based methods have shown promising results in automated DR detection using fundus images. However, many real-world fundus images suffer from low-light conditions, noise, and poor contrast, which significantly degrade model performance. Moreover, most existing deep learning models are computationally expensive and unsuitable for deployment on resource-constrained edge or IoT devices. These limitations reduce the practical applicability of automated DR screening systems in real clinical and rural health-care environments.

3. Research Gap and rationale of the research project

Existing research on the detection of diabetic retinopathy focuses primarily on achieving high classification accuracy using large deep learning models under ideal imaging conditions. Limited attention has been paid to the handling of low-light fundus images, which are common in real-world screening scenarios. In addition, most approaches do not consider model uncertainty, which is critical for safe medical decision-making. Furthermore, the deployment of such models on edge devices remains challenging due to high computational and memory requirements. Therefore, there is a research gap in developing a lightweight, uncertainty-aware DR detection framework that can effectively handle low-light images while being suitable for edge deployment. This study aims to address these gaps by combining diffusion-based image enhancement, knowledge distillation, and efficient transformer-based architectures.

4. Research Questions

- How can low-light fundus images be enhanced to improve diabetic retinopathy detection performance?
- Can knowledge distillation enable a lightweight student model to achieve performance comparable to a high-capacity teacher model?
- How can uncertainty estimation be incorporated to improve the reliability of DR detection systems?
- Is the proposed framework suitable for deployment on resource-constrained edge devices?

5. Objective of the Research

The main objective of this research is to develop an efficient and reliable diabetic retinopathy detection system that can operate under low-light conditions and be deployed on edge devices.

5.1. Specific Objectives

- Enhance low-light fundus images using diffusion-based image enhancement techniques.
- Develop a high-capacity teacher model for the classification of diabetic retinopathy.
- Design a lightweight student model using knowledge distillation and adapter-based learning.
- To incorporate uncertainty estimation for reliable medical decision support.
- To evaluate the proposed model on multiple public DR datasets.
- To optimize the model for deployment on edge and IoT devices.

6. Literature Review

Several studies have explored deep learning-based approaches for automated detection of diabetic retinopathy using convolutional neural networks and transformer-based models.

Recent work shows that vision transformers can capture global context and improve classification performance. Other studies focus on image enhancement techniques to improve fundus image quality, including low-light and noise reduction methods. Diffusion models have recently gained attention for high-quality image restoration tasks in medical imaging. Knowledge distillation has been widely used to transfer knowledge from large models to smaller models for efficient deployment. However, there is limited research that jointly considers low-light image enhancement, uncertainty estimation, and edge deployment for DR detection.

7. Research Methodology

Figure 7.1 illustrates the overall workflow of the proposed diabetic retinopathy detection framework, including preprocessing, diffusion-based low-light enhancement, knowledge distillation, and uncertainty-aware prediction

7.1. Sources of the Data

In this study, datasets of publicly available diabetic retinopathy such as EyePACS, APTOS, and Messidor will be used.

7.2. Intended Population and Sampling

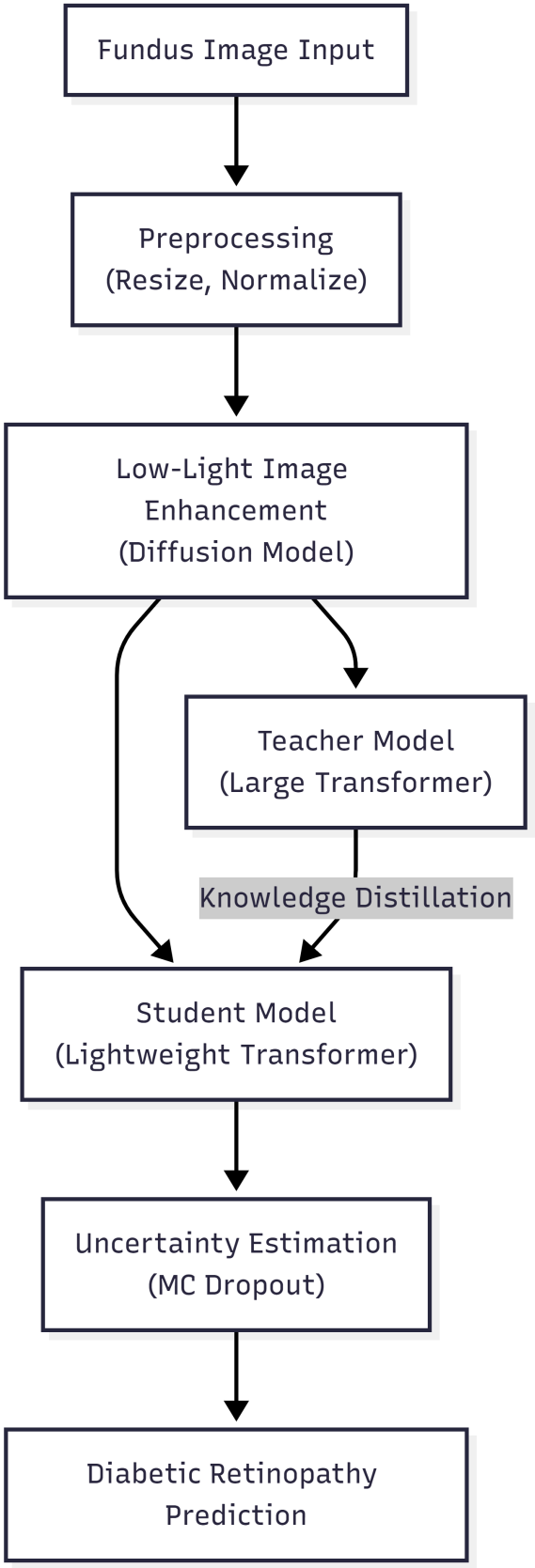
The intended population includes fundus images of diabetic patients across different stages of diabetic retinopathy. Dataset splits will be performed to ensure proper training, validation, and testing.

7.3. Data Collection Technique

The study will use secondary data obtained from public datasets. Images will be preprocessed through resizing, normalization, and low-light simulation techniques.

7.4. Data Analysis Technique

A diffusion-based model will be used for image enhancement, followed by training a teacher model and a lightweight student model using knowledge distillation. Model performance will be evaluated using standard classification and uncertainty metrics.



Proposed Methodology of this study

8. Expected Results of the Study

The expected outcome of this study is an improved diabetic retinopathy detection system that performs robustly under low-light conditions. The proposed student model is expected to achieve competitive performance while being computationally efficient and suitable for edge deployment.

9. Key Performance Indicators

Accuracy, F1-score, AUC-ROC, model size, inference latency, and uncertainty calibration metrics such as Expected Calibration Error (ECE) will be used to evaluate the proposed framework.

References